



▲ SHRIMP POOLS

The Far East has a long tradition of mariculture, and for centuries, fish and other marine animals such as tiger shrimp have been raised in tidal pools dug for the purpose on low-lying coasts. Unlike normal fishing this is a form of farming rather than hunting, so it has no direct effect on the wild stocks. But it can have other environmental impacts, especially when tidal mangroves are cleared to make way for the pools. This destroys habitats that are a vital refuge for the young of many valuable sea fish, and also makes local communities more vulnerable to oceanic storms.



▲ MUSSEL BEDS

Mussels and oysters are ideal for farming, because they naturally attach themselves to rocks and other hard surfaces and do not bury themselves in the sand or mud. These mussels are being grown on ropes twisted around wooden posts on the Atlantic coast of France, where they are submerged by the tide twice a day.

fishing

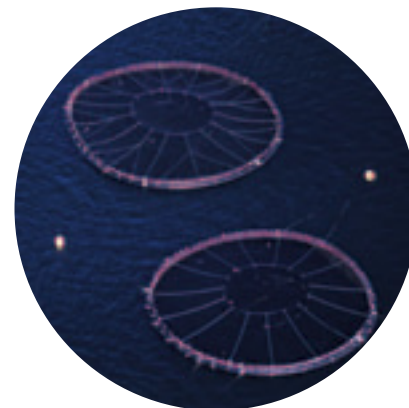
SATELLITES AND FISHFINDERS



Fishing skippers are aided in their search by a variety of high-technology equipment. Many have access to satellite imagery that shows where dense plankton swarms are likely to attract fish. They also use refined echo-sounders called fishfinders that use sound pulses to detect fish in the water below the ship. These can penetrate as deep as the twilight zone, and reveal fish shoals as distinctive traces on the screen called "fish arches." Several of these can be seen on the fishfinder screen shown above, along with a broad band of white that indicates the position of the seabed.

► SALMON CAGES

Most sea fish do not do well if they are kept in confined spaces, but salmon are an exception. They can be raised in submerged cages like these, in clear, cool tidal waters that sweep through the cages and keep the fish healthy. They have to be fed, however, and the uneaten food and fish waste can upset the local balance of nature. Despite this, the salmon farming industry is very successful, and has reduced fishing pressure on wild salmon.



► SEAWEED FARMS

Seaweed is a surprisingly valuable material, since it is used as a thickening agent in a wide range of products, including toothpaste and ice cream. It is also cooked and eaten in many areas, especially in China and other countries in the Far East. A lot is gathered from the wild, but in the Indo-Pacific region seaweed farming is a flourishing industry, with an annual production that exceeds the output of all other forms of mariculture put together. Here, a seaweed farmer tends her crop in the shallow waters off the coast of Zanzibar, east Africa.

